

The Logic of the Fundamental Ideas

in Brief Compass.

By

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Foreword

In this little book the author has endeavoured to give a short and clear presentation of his theistic logic. He has therein found occasion to bring out the nodal points on which the scientific dispute between pantheism and theism must essentially turn.

R. Nielsen.

Introduction

At the present stage of philosophy's development it becomes a scientific necessity to treat logic from three essentially different points of view.

In order to develop the laws of thinking, one must keep to thinking itself. But thinking is a human activity of spirit; we must learn what thinking is from our own thinking. It lies near, then, to regard the human being as the thinking subject, to reflect upon the mutual relations of the acts of thought, and to mark the forms necessary for the understanding, the intellectual traces which advancing thought deposits. This point of view is characteristic of *formal logic*.

It is, however, so far from being the case that the finite understanding should command the demands of thought, that these must on the contrary be said to command the thinker; there shall be thinking, but when the thinking is consistent it is for the rest indifferent who it is that thinks — whether it be a human being or a god or perhaps pure being. The emphasis falls on the thinking. For thinking's own sake, on the other hand, it is of importance to fix the eye on that which is thought, on the universally necessary content; for the content of thought and the form of thought are determined by one another. This point of view is characteristic of the *speculative* or, to use a perhaps still more telling expression, *the immanent logic*.

These two principal forms of logic, of which the first has an Aristotle, the second a Hegel, as supreme representative, may with reason be said to enjoy a well-deserved recognition in science.

That they cannot well be reconciled with one another, but stand in a most serious dispute concerning the very first principles of thinking, notably the principle of contradiction, is certainly less commendable; but they will both maintain their place:

the one will not be able to displace the other.

In order that each of the contending parties may come fully into its right, it is in our judgment necessary that yet a third principal form be introduced. But before we attempt to prove the correctness of this assertion, we consider it expedient to bring the architectonic opposition between the two doctrinal edifices nearer to intuition by prefacing an outline corresponding to each of them.

Formal Logic.

Formal logic, or the doctrine of the abstractly necessary thought-forms, divides into the doctrine of *Concept*, *Judgment* and *Inference*.

Concept.

Thinking, which is already present in sensation, raises itself forth as it dissolves the immediate sense-images, forms by comparison and abstraction a *general image*, holds it fast in the *word*, makes clear to itself the essential marks that necessarily belong together, and transforms it into a *concept* (concept-analysis, definition).

The sum of the determinations belonging to a concept constitutes the concept's *content*; the number of the objects to which it finds application designates its *extension*. (Inverse relation between content and extension; superordinate and subordinate concepts; genus- and species-concepts.)

Judgment.

The concept finds its pure logical expression only in language: words are joined into sentences, and in sentences judgments are uttered. The sentence whose form expresses a concept is a *logical judgment*. In order to express the concept in this form, the judgment must formally indicate: the *Subject*, the representation that is to be determined; the *Predicate*, the determining representation; and the *Copula*, the determining word of connection.

From the subject's connection with the predicate, determined by the copula, it follows that the judgment can be separated into a plurality of logically different judgments, and judgments are divided: according to *quality* into affirmative, negative and infinite; according to *quantity* into singular, particular and general; according to *relation* into categorical, hypothetical and disjunctive; according to *modality* into assertoric, problematic and apodictic.

If the predicate proceeds from the subject with universal logical necessity, the judgment is *analytic*; if the predicate can be joined to the subject only through a particular intuition or experience, the judgment is *synthetic* (synthetic judgments *a priori*; Kant's problem).

Inference.

The judgment must be grounded; its validity must be seen into; the act of thinking whereby the validity of one judgment is derived from one or more other judgments is a *logical inference*. If the inference has only one antecedent proposition (premise), it is called *immediate*; if it has several premises, it is called *mediate* (the requisites of inference).

The mediate inferences are divided into: the *categorical*, conditioned by the subsumption of the subordinate concept's mark (*nota notæ*) under the superordinate (syllogistic figures); the *hypothetical*, which rest on the dependence of the conditioned upon its condition (*modus ponens* and *modus tollens*); and the *disjunctive*, which are grounded on this, that the members of a whole are mutually so related that when one is expressly posited the rest are excluded (*modus ponendo tollens*), and that when the rest are excluded the one remaining must be posited (*modus tollendo ponens*).

By treating the doctrine of proof and method, formal logic resolves itself into a general doctrine of knowledge and of science.

The Immanent Logic.

The immanent logic, which is a dialectical development of the system of the categories (the thought-forms inherent in all being), divides into the doctrine of *Being*, *Essence*

and *Concept*.

Being.

Universal being is one with universal thinking; but the unity of thinking and being is, in its most abstract form, void of content: pure *Being* is the same as pure *Nothing*. But *Nothing*, as non-being, is nevertheless different from *Being*. That *Being* both is and is not *Nothing* means that *Being* unceasingly passes over into *Nothing*, and *Nothing* into *Being*: this transition is *Becoming*.

In *Becoming*, *Nothing* is a limit for *Being*, and *Being* for *Nothing*. *Being* limited by its *Nothing* is a *being*, a *Something*, and *Nothing* determined by *Being*, an *Other*. But the *Other* is itself *Something*, and *Something* in turn *Other*; *Something* passes by its own limit over into *Other*, and *Other* into *Something*: this transition is *Alteration*.

In *alteration* *Something* is one with its *Other*; this is the *Unchangeable* in the *One*; but *Something* is also different from the *Other*; this is the *Changeable* in the *One*. It is thus the *Unchangeable* that is altered; but precisely by being altered it keeps itself unaltered: under all *alteration* there is something *unchangeable*.

Essence.

The *Unchangeable* that is seen in the *Changeable* is *Essence*; the *Changeable* through which the *Essence* is seen is *Semblance*: the alternating shine of *Semblance* and *Essence* is *Reflection*.

In its identity with the *Essence*, the *Semblance* is *Phenomenon*; in its difference from the *Phenomenon*, the *Essence* is *Ground*. The *Phenomenon* goes to the ground; it belongs to *Semblance* and must be reflected into its *Essence*. But in that the *Phenomenon* receives the *Essence* for its ground, it becomes grounded and is posited as *Existence*.

Existence reflected into itself is an *existent*, a *Thing*; but the *Thing* can be reflected into itself only by reflecting its existence into other existing things: the determinations corresponding to these reflections are *Properties*.

In this mutual reflecting the things dissolve into the properties, and everything

becomes unstable. But the unstable presupposes something subsisting which cannot be dissolved, because its subsistence infinitely confirms itself in the alternation of the unstable; this subsisting something is the *Substance*, and the modes of appearance alternating in instability are the *Accidents*.

The relation between the Substance and its Accidents is a *relation of causality*: the Substance is *Cause*, the original matter; the Accidents are *Effects*. But the Cause has its substantial expression in the Effect; without Effect, no Cause: by making the Cause a Cause, the Effect is itself Substance — thus a Cause that makes the Cause an Effect.

In *Reciprocity* the Effect is Cause, and the Cause Effect; and in that all the moments of actuality contained in the Essence are, under all-sided reciprocity, developed in full totality, the Essence developed in the totality is Concept.

Concept.

The self-comprehending totality is, in its unity with itself, the *Universal*; in its difference from itself, *the Particular*; in its totalized determinateness through itself, the *Singular*.

The Concept's self-development is its *ground-division* (the Judgment). By virtue of the ground-division, the totality in its first immediate being, the Singular, is posited as the *Subject*, and the totality in the reflected form, the Universal, as the *Predicate*: the Subject is what the Predicate is; the *Singular* is the *Universal*.

But in that the Singular is comprehended as the Universal, and the Universal as the Singular, the opposed members of the concept-separation become mutually comprehended, in reciprocal determination, as the *Particular*.

The outer members of the ground-division, the *Extremes*, have thus in the Concept's self-development deposited a *middle member*, a *Medium*, in which they can be mediated. This mediation is the *Inference*.

The Concept's self-mediation through Judgment and Inference is its *Subjectivity*; the immediacy regained through the mediation is its *Objectivity*.

As immediate, existing totality the Concept is Object — and, by virtue of the ground-division, an infinite manifold of existing objects.

The objects in their outward, indifferent mutuality are *mechanical*, acting upon

one another by *pressure* and *impact*. The separation of motion and rest which herewith enters is a relative appearing of the *absolute Mechanics*, in which motion is rest, and rest motion.

Beneath the indifference of the mechanical objects there hides a *difference*, which comes to light in chemical attraction. The difference of the objects brings with it a tension, which is equalized in *the chemical product*. But the product, in which the difference is neutralized, must as the *Neutral* dissolve again into difference, and the *process* begins anew.

In the chemical process the Concept goes forth out of its own presuppositions, sublates the presuppositions in their result, in order again to sublate the result in the presuppositions. The objective Concept is now master over the material: the result anticipated in the presuppositions is *End*, the presuppositions are *Means*, the movement is *teleological*.

The teleological Concept, comprehending itself in its own elements, is a form that gives itself content, a content that gives itself form: in this its *absolute unity* of form and content, of *Subjective and Objective*, of end and means, the Concept is *Idea*.

From the outlines here set forth it will be seen how divergent the directions are. In formal logic the human being, the thinking subject, is from first to last occupied with his own acts of thought; it is the human being himself who abstracts, reflects, judges and infers. But for just that reason the thought-forms are empty forms, and the acts of thought finite, psychologically conditioned operations of the understanding. The object of thinking falls outside the thinking. Here is an insurmountable gulf between the subjective and the objective, and the thinker has only to watch that in his judgment about the things he does not come into contradiction with himself. To secure the unity of the thinkable and the possible, one holds fast to *the principle of contradiction*; but that the principle of contradiction affords no reliable guarantee of the objective validity of subjective thinking, the *antinomies* in Kant's Critique of Pure Reason have sufficiently shown.

The gulf between the subjective and the objective disappears in the immanent logic, which abstracts from the thinking subjectivity and lets Being think itself by the help of contradiction and negation. That Being is Nothing is indeed a contradiction, for Being

must necessarily exclude non-being. But the non-being that is excluded is precisely the negation that lies in Being's own concept. That the Unchangeable is changeable, that the Essence is Semblance, that the Cause is Effect, are sheer contradictions. But these contradictions make it precisely evident how the thinking that is entangled in finite representations deceives itself by keeping the content of what is outside itself. The Changeable is the Unchangeable's indwelling negation, just as the Semblance is the Essence's, and the Effect the Cause's.

Nevertheless it must not be overlooked that the immanent method of developing the matter, despite its epoch-making significance, also has its weak sides. Negation is originally a thought-determination, not a determination of being. What is can, in its immediacy, negate neither itself nor another being. Negations, contradictions and resolutions of contradictions are acts of thought which hold of what is only under the express condition that Being is thought. But in order that this condition can be fulfilled, the invoked unity of thinking and being must from the beginning be grounded upon a corresponding opposition. Pure Being is Nothing, when Being is thought; but Nothing and Being fall nonetheless each under its own sphere: Nothing under thinking, and so under the subjective; and Being under actuality, and so under the objective.

If the dispute between the two recognized points of view is not to become endless and unfruitful, a third point of view must be chosen, from which it becomes possible to reconcile the contending parties. It must then be conceded to formal logic that the thoughts proceed from a thinking subjectivity; but it must at the same time be conceded to the immanent logic that it is not a finite but an infinite thinking that comprehends existence. Formal logic must be right in this, that there is a characteristic difference between the subjective in thinking and the objective in actuality; but the logic of immanence retains its right in this, that the opposition of the subjective and the objective rests upon identity of thinking and being. It is this third point of view that is chosen in *the Logic of the Fundamental Ideas*.

As regards the origin of thoughts, only three assumptions are possible: 1) that they proceed from a finite subjectivity; 2) that they develop themselves out of universal being; 3) that they spring from an infinite, absolute subjectivity. As regards the relation between thinking and being, there are likewise only three assumptions possible: 1) that

it is a finite opposition, be it in agreement or in discord; 2) that the unity sublates the opposition; 3) that the unity grounds the opposition.

For the all-sided development of logic the three points of view here set up are necessary; but more than these three are, so far as principles are concerned, impossible.

The Logic of the Fundamental Ideas

develops the ideal essentiality of the thought-forms under *the Idea of Knowledge*, the forms of being corresponding to actuality under *the Idea of Power*, and points out the dualistic unity of the ideal and the real content in *the Idea of Truth*.

The fundamental separation of the three Ideas is especially motivated by regard to the one-sidedness of the immanent logic.

In order that thinking shall be possible, there must be something that is thought, hence something that thinking itself posits, presupposes and produces; this is *the Ideal*. But in order that actuality, the actual world, can have validity, there must be something other than thinking: something that is not thought, but about which there is thinking; something whereby thinking is determined, but which it does not itself produce; this is *the Real*. The difference between the Ideal and the Real points to a decisive difference between Knowledge and Power: the Ideal is a knowledge-determination, the Real a power-determination.

It is universally acknowledged that everything in the world must have a ground; herein lies that the existent must rest upon a non-existent. The ground by itself is not existent; it has existence only in the existence which it grounds. The ground is an *ideality*; the thing, the actual object, is a *reality*. The ideality can be thought, and there can be thinking about the reality; but the reality itself cannot be thought; to think the reality is to sublate it and transform it into ideality.

The immanent logic has shown excellently what ideality is, but it has entirely forgotten what reality is.

When the finite, the existing positive, is to hold itself abstractly, as the Real, over

against the infinite as the negative, the Ideal, then it is easy to see that the ideality must swallow up the reality. For by having the infinite as its limit, the finite itself becomes infinite; but thereby the infinite, which from its side receives the finite as its limit, becomes precisely finite. In that the opposed qualities have thus mutually negated one another, they have, by the negation of the negation, at the same time affirmed one another, and have thereby both obtained the same ideality: the finite's ideality is its sublation and confirmation in the infinite; just so the infinite's ideality is its sublation and confirmation in the finite. But the reality has vanished.

The immanent logic has an altogether false conception of reality. In its zeal to comprehend the object, it neglects to form an actual concept of the actual object. The actual object is not an ideality but a reality, a phenomenon of power, which no thinking is able to produce. Not even the divine thought would be capable of thinking an actual stone; for when the stone is thought, it is transformed into ideality and ceases to exist.

It is with respect to this that the Logic of the Fundamental Ideas has enforced the dualistic opposition of Knowledge and Power. If Power is to pass straightway into Knowledge, then it is over with the reality of existence; the things become semblance, and the actual world dissolves into a metaphysical system of divine thoughts. If Knowledge is to vanish into Power, then ideality is lost. Realism sinks down into a crude materialism, which makes the Ideal a semblance of the Real, the spiritual an empty after-glow of the corporeal.

But when all the ideal content of being is elucidated in *the Idea of Knowledge*, and the real content of actuality passes into *the Idea of Power*, then it follows of itself that the unity of the opposition must have its complete expression in the Idea of Ideas, the all-sided Idea, which saturates thinking with ideality and upholds the reality of actuality — i.e. in *the Idea of Truth*.

A. Knowledge.

a). The Concept of Knowledge.

To know something is, in indeterminate generality, to become conscious of something. Here, then, are presupposed¹: *a knowing self, an object of knowledge, and consciousness's unity of both*. In the apprehension of this unity what chiefly matters is certainty, assurance, and insight. Knowledge is the intellectual light in which selfhood-determinations and determinations of the other are united and clarified in full transparency.

α) Immediate Certainty.

Through sensing and thinking the manifold content of existence passes over into consciousness. But since thinking in its beginning is just as immediate as sensing, self-certainty and certainty of the other are originally, though in different ways, immediate.

Immediate certainty is groundless, hence unfortified, and passes involuntarily over into uncertainty. The objects alter: doubt arises about the essence of the objects. The senses can deceive; doubt arises about the testimony of the senses. Thinking develops concepts; but the partially formed concepts stand in contradiction with one another:

¹As regards the determination of philosophy's first general presupposition, one must indeed distinguish between the propaedeutic preparation and the systematic beginning. In the philosophical propaedeutic the relation between the knower and the object of knowledge is elucidated from the ever more concrete standpoints of the doctrine of cognition, the doctrine of science, and the doctrine of Ideas. (Cf. *Philosophisk Propædeutik i Grundtræk*. Kbhvn. 1857). In logic, by contrast, the point of departure is taken in pure, self-presupposing knowledge. (Cf. *Grundideernes Logik. I*. Kbhvn. 1864. Side 1–11).

doubt arises about the validity of thinking (scepticism²).

In order that certainty may be won back through uncertainty, there is required

β) Assurance.

On the basis of immediate certainty, self-certainty was presupposed without further ado, and there developed, in the clear light of immediacy, a manifold knowledge of the other (the world of objects and representations). But doubt came and darkened the clarity. Doubt, which dissolves knowledge and confuses consciousness, begins by undermining the certainty of the world of objects and ends by attacking self-certainty. But in its attack on self-certainty doubt dissolves itself and perishes in renewed certainty. Even if the doubter could doubt everything (*de omnibus dubitandum est*), he could nevertheless not possibly doubt his own doubting. Doubt is thinking, and the self is thinking (*cogito ergo sum*); it is, in its self-thinking, assured of its own being. But thereby it becomes at the same time assured of the being of things.

From self-assurance follows the assurance of the subjective validity of the original selfhood-forms (the pure intuitions, categories, and Ideas) (Kant); but from this follows also the assurance of their objective validity. If there could be otherness-forms, determinations of being in the thing in itself (das Ding an sich), which by their nature stood in conflict with the original selfhood-forms, then it would be not merely impossible for us, but impossible in itself, to have a knowledge of the essence of things; for not a single determination can be taken up into knowledge without the knowing self having to produce it. This holds not only of thoughts but also of sense-impressions, not only of concepts but also of images and representations.

Judging by immediate experience, one would indeed suppose that from the world of things there were conveyed to consciousness countless object-forms and property-determinations which did not in the least concern its own essence. Yet upon some reflection it will readily appear that nothing can be known except that which is determined in knowledge. How should there be a knowledge of colour if the knowing being did not itself have something to do with colour.

²On the sceptical dialectic cf. Phil. Propæd. Side 152–154; Grundid. Log. II. 212–222.

To appeal to visual organs and light-impressions is in this connection of no significance, for the fundamental question is how the organically conditioned light-impression is transformed into consciousness of light, or how a colour-determination can become a knowledge-determination. Could the light from without shine through the organ into the soul, as through a prism it can fall upon the wall of the chamber, then the colours would indeed be able to colour consciousness, so that it became both yellow and blue.

γ) Insight.

Through assurance the lost clarity is regained in the form of clearing-up, grounding, and explanation, and corresponds to insight. The knower's insight is double: it is an insight into the matter with its members and relations, and it is an insight into the proper essence of the thinking consciousness. Just as thinking of the other is essentially self-thinking, so all thorough understanding of the other is conditioned by self-understanding. Logical self-understanding rests on the insight that the forms of reflection, judgment, and inference are universally valid otherness-forms because they are original and necessary selfhood-forms.

The knowing subjectivity is in its self-knowledge a reflecting being. Immediately, reflecting is a mirroring, and presupposes a mirroring activity and a mirror-image: pure knowledge is, in the intellectual sense, pure mirroring. That the subjective activity is mirroring makes it explicable that what is mirrored can remain standing in the form of otherness without melting together with the determining selfhood-form — that colour-images, for example, can be mirrored by consciousness and awaken consciousness of colours without colouring consciousness. But so long as mirroring is apprehended only as immediate reflecting, this reflecting will fit otherness just as well as selfhood. In optics there are many mirrors and many kinds of mirrorings; but logically understood, the original reflecting is an ideal self-activity, a thinking. Things cannot reflect, for they cannot think; on the other hand, not only can their essence be thought upon, but this essence can be reflected in itself when it is reflected in subjectivity: all reflection of the other rests on self-reflection.

The knowing subjectivity is in its self-knowledge a judgment; herein lies the

ground of the judgment's form being an essential otherness-form. Subjectivity is a judgment, that is, it is a subject whose predicates are determined by the functions corresponding to its judgments. "The self is seeing, is hearing, is thinking"; by the predicate "seeing" are determined all judgments about visible things, by the predicate "hearing" about audible ones, by the predicate "thinking" about thinkable ones. From this it follows that in every possible judgment, whether the self for the rest judges about the other or about itself, there must be a double subject, the judgment's and the judger's, and likewise a double predicate and a double copula: the otherness-form in the judgment is infinitely reflected in the selfhood-form.

The knowing subjectivity is in its self-knowledge an inference; herein lies the ground of the inference-form being an essential otherness-form. The inference arises — so far as one looks to what is typical — in that the judgment's subject and predicate, according to their difference of totality, form extremes whose unity cannot immediately be indicated by the copula but must be mediated by a middle term.

The selfhood-judgments declare, namely, that heterogeneous predicates, belonging to different spheres, are posited in one and the same subject. In seeing, the subject is the seer; in hearing, it is the hearer; the seer is thus hearing, and the hearer seeing; the sensing one is thinking, and the thinking one sensing. Raised above the difference of the knowledge-functions, the self holds itself fast as the singular. And yet the self is in itself a particular, for sensing is something other than thinking; what is thought is not sensed: the sensing one is thus not the thinking one, or: the sensing and the thinking self are particular. Further, sight is something other than hearing; what is seen cannot be heard, and what is heard cannot be seen: the seer is thus not the hearer, or: the seeing and the hearing self are particular.

Now by such a division the self would be split asunder if the division were not a self-division that let itself be equalized in the universal. But the self is itself this equalization.

The self divides itself into its functions; but these functions, however heterogeneous they may be, are yet all selfhood-functions, and, as selfhood-functions, functions of one another mutually.

If in sensing there were not a hidden thinking, the sensing self could distinguish

nothing, and consequently could not sense: sensing is thus a function of thinking. Conversely, if in thinking there were not an implied sensing, there would be no element of representation, and the thinking self could distinguish nothing, and consequently could not think: thinking is thus a function of sensing. One distinguishes in sensing by the help of thinking, and one distinguishes in thinking by the help of sensing. That there is distinguishing is the universal; that which under the universal distinguishing is separated and distinguished is the particular. The distinguisher is in its distinguishing equal to itself: the self that distinguishes itself in everything and everything in itself is thus the particular because it is the universal.

Apprehended immediately for itself, the self is the singular. In this its singularity it is a fixed point, an atom, from which activities go forth in all directions. To this attaches the representation of an existent thing, a soul-real (Herbart); but such a representation makes the self selfless.

Apprehended immediately in the form of particularity, the self is a swarm of selfhood-points: it is the centre of the whole circle, but every point in the circle is in turn a centre. This monstrous contradiction stems not from particularity, but from the immediacy in which the particular is held fast. The dissolution of immediacy in infinite reflection is the solution of the contradiction. The self is reflected in its singularity, for its activity as a singular is, by means of the particular, a self-activity. It is reflected in its particularity, for its particular activity is, by means of its singularity, a self-activity: self-activity = self-activity is the universal in which the self consists.

Knowledge is not activity, but the light in the universal activity, the infinite reflecting and mirroring. The universal activity, an activity of activity into the infinite, is in its formal generality an empty concept. The essence that is universal in the absolute sense is thus self-essence only in that the singular and the particular, each for itself in mutual opposition, is self-essence.

Where then is the self in this essence? It is in its own inference. When the singular and the particular are posited as extremes, the middle is the universal (S—U—P); when the singular and the universal are posited as extremes, the middle is the particular (S—P—U); when finally the particular and the universal are posited as extremes, the middle is the singular (U—S—P). The self has subsistence in a system of inferences:

the inference-unity, the absolute central unity, is itself the eternal beginning-unity.

From this standpoint it will now be possible to attain insight into the relation between the logical fundamental forms of selfhood and of otherness.

In the essence of knowledge they are the same forms; the opposition rests solely on reflection's position toward the immediate. In the selfhood-forms immediacy is completely dissolved; but in the forms of otherness the immediate constantly returns and interrupts reflection. Otherness cannot reflect itself; it must be reflected by selfhood.

As object for the self, that which is is itself a something for itself. That it is something for itself points to selfhood; and yet something is for itself only what it is in its relation to an other. But something passes over into an other only so far as it is thought together with the other; its transition is no self-movement — the movement is in the thinking subjectivity. Something reflects itself in an other only in that the self reflects upon something and other: in becoming and alteration the otherness-reflection shows itself expressly as the intellectual product and counter-image of self-reflection.

Otherness-being is becoming. If that which is is reflected in its own becoming, and conversely: then immediacy is sublated; then that which is is essence, and that which becomes, semblance. As essence and semblance reflect themselves in one another, it looks as though the essence itself thought its semblance, and the semblance its essence; but the thought-movement goes on in the thinker. The reflection of semblance and essence is self-reflection's own reflection of the other. That the reflection-form, essential in itself, is an otherness-form is seen also from this, that the essence's reflection into itself is posited as thing, and its reflection into an other as property: the thing itself is selfless, and the property a quality of the selfless.

But the selfless must necessarily acquire a semblance of selfhood; otherwise the otherness-forms could not possibly become knowledge-forms. The knowing self then knows that the concept of the thing's being is a judgment. The thing now becomes subject, and the property its predicate. The subject is its predicate: it looks in otherness quite as in selfhood. But in the essence of otherness the selfhood is semblance; the thing, the selfless subject, has no true subjectivity; the subjectivity lies on the surface, for it lies in the properties.

By making the world of things into a world of judgments, the judging self knows how to hold things together and to comprehend the unity of connection in a system of inferences. Unity of connection is determined by relation, relation by reflection, reflection by judgment, judgment by inference: the articulated reciprocal determinations of the forms of selfhood and otherness yield the fundamental moments in the concept of knowledge.

b) Finite Knowledge.

According to the concept of knowledge, self-knowledge and knowledge of the other must necessarily presuppose each other into the infinite; but in the special development they part company, and herein finitude is seen. The distinguishing mark is that the knowledge-content which saturates the selfhood-forms and is necessary for their logical unfolding is poor, empty, and formal in comparison with the content that fills the forms of otherness with their outwardly infinite wealth. The strict, systematic deduction of the original I-hood propositions, as it has been carried through in subjective idealism (Fichte), is a telling example of the a priori poverty of pure self-knowledge. In contrast, the doctrine of experience has such a superabundance of content-determinations and objective knowledge-forms that along this path there can be no question of demonstrating the necessity with which the facts of the objective world proceed, member by member, from the subjective fundamental demands of the I-essence (nature-philosophical mysticism; Schelling).

In order to make the relation of finitude clear, we note especially: first the one-sidedness of a priori knowledge, then that of empirical knowledge, and thereby come to see the necessity of the logical dissolution of finite knowledge.

α) A priori Knowledge.

Formal logic's conception of the principle of identity and contradiction finds its most adequate application in the science that is precisely suited to make intelligible the finitude of a priori knowledge, namely mathematics. The equation: $A = A$, is in the objective sense what the self-proposition: $I = I$, is in the subjective; likewise the

proposition: A is not Not-A, in the form of otherness, has as its presupposition the proposition: I am not Not-I, in the form of self-limitation. Here is immanent reflection of the subjective and the objective.

Further, a hypothetical proposition such as this — if $\frac{a}{b} = \frac{c}{x}$, then $x = \frac{bc}{a}$ — can serve as an example that objective relation is grounded in subjective reflection. From the antecedent the consequent must necessarily follow; the relation between a, b, c, and x is so far objective. But the magnitudes have not themselves set themselves in relation to one another; a has not itself divided itself by b, nor c by x. The relation is posited by the reflecting subjectivity. In the quotients $\frac{a}{b}$ and $\frac{c}{x}$, a is reflected in b, and b in a, c in x, and x in c; the numerator presupposes a looking to the denominator, and the denominator to the numerator: that is subjective reflection. If an equation such as $\frac{dy}{dx} = c$ is chosen, the numerator is no immediate term, but a reflex of a magnitude vanished in vanishing, and the same holds of the denominator dx; as the two reflexes are reflected in each other and form the quotient, the reflection becomes objective; but it is only objective because it is subjective.

Could the mathematical development now be continued in such wise that to every new objective relation there corresponded a new and fruitful insight into subjective reflection, to every new theorem a new and fruitful cognition of subjectivity's original self-positing: then we should here have a method (*methodus mathematica*) which was in an eminent degree suited to sublimate the finitude of a priori knowledge.

But this reckoning does not suffice. It is true, indeed, that propositions such as, A is A, A is not Not-A, belong so intimately to general self-thinking that their validity is evident to every thinker. But whence comes it that equations such as $\log A^x = x \log A$, d. $\sqrt{a^2 - x^2} = \frac{-x dx}{\sqrt{a^2 - x^2}}$, even when the notations are explained one by one, cannot be said to belong to the original comprehension of the self, are not indispensable for logical self-understanding? Nay, whence comes it that mathematics cannot be deduced from metaphysics, or conversely? It comes from this, that a priori knowledge, on account of finitude, parts into two one-sided directions: a logical-dialectical one, which immerses itself in self-reflections, and an exact-operating one, which loses itself in reflections of the other³.

³On the difference between reflection-concepts and operation-concepts cf. *Mathematik og Dialektik*. Kjøbenhavn. 1859. S. 40—56.

β) Empirical Knowledge.

The a priori otherness-forms that are articulated in the pure analyses of magnitude rest solely on phantasy-intuitions and representations, and the represented otherness falls, so far, still within selfhood.

But when it is a question of a knowledge of existing things, the contradiction enters that the objects within knowledge themselves fall outside knowledge. The finite consciousness has, namely, a world of objects *outside* itself; but it cannot possibly have a knowledge of these objects except by taking their being up into itself, and thus having them *within* itself. This contradiction, hidden in the foundation of empirical knowledge, conditions the manifoldness of cognition, but at the same time betrays its finitude under the form of contingency.

The contingency is subjective, for the universally necessary otherness-forms in which the objects must be apprehended can indeed be had in consciousness without knowledge that the special objects really exist. That an existent object must always correspond to some concept or other, that is, to some filling-out or other of a priori forms, does not sublate the contingency, since such a filling-out can take place in countless ways: that many and different things bear the same name points precisely to the differences being regarded as contingent. The empirical concept to which the name answers repeats itself in a multitude of corresponding specimens; on the basis of the concept the specimen is described, but the pictorial elements of the description indicate the contingency.

In relation to the individuals and the immediately individual, the existent concept is the species, and the nearest universal concept of the particular species is the genus. That an individual with one or more marks of a certain species must have all the marks belonging to the species is inferred by *analogy*; that essential determinations which are decisive for some individuals belonging to a certain species or genus must hold for them all is inferred by *induction*; but since not only the individuals but also the concepts are empirical, both induction and analogy are burdened with a moment of contingency that can transform certainty into uncertainty.

From a general standpoint empirical knowledge can indeed attain reliable certainty; but the particular cases constantly cross the universal and call forth uncertainty. The

ground of assurance is negative: “here is no case that breaks with the universal established by the observations; ergo this universal, determining the cases, is a law.” But the universal thus established is nevertheless, in its positive form, empirical. To the methodically continued observations the world of experience constantly presents new sides, new relations, new properties, new laws.

The contingency is objective, for it is not merely we who must alter, correct, and improve our concepts of things; the concepts themselves (species and genera; Darwin) alter by degrees, in that, as existence-concepts, they alter the existential expression with the alterations of the individuals. The contingency is objective, for the existence of the objects is independent of our subjective knowledge of them. We could not sense the objects if the objects did not sensibly exist; but the sensible objects can very well exist without our sensing them. We form for ourselves concepts that are abstracted from the objects, but the objects themselves are indifferent to our abstracting and comprehending. It lies in the independence of the objects to have our knowledge of them outside themselves as a contingency.

For our finite knowledge, then, the world of things stands as a barrier, an impenetrable other. We cognize, says the doctrine of experience, the properties of things, but not the things themselves; we cognize the laws, but not the essence. Mechanics does not know what force is; chemical physics does not know what electricity is; physiology does not know what life is. True empirical knowledge has everywhere taken up into its foundations the determination: not-knowledge.

γ) The Dissolution of Finite Knowledge.

The contradictions with which finite knowledge is burdened on all sides gather at last into the one: omniscience is necessary and yet incomprehensible. That omniscience is necessary shines forth from the original essence of knowledge. For since all determinations of existence are, in the logical sense, knowledge-determinations, and all knowledge-determinations functions of the knowing self: nothing whatever can be known except that alone which is produced by knowledge.

The contradiction is now this, that finite knowledge everywhere in the actual world finds before it object-forms which neither it itself nor the object has produced.

All outward things exist in space. But what is space? An extension produced by intuiting, existing for intuition. In intuiting the things in space, we intuit them in a previously intuited form (objective space); and the same holds of time.

By reflecting on the objects existing in space and time, we come to a similar result. For us, with our finite knowledge, the intuitable objects exist before we intuit them. But intuitable is only that which is produced by intuiting. Our intuiting is determined by the intuitable, that which is already produced. How has the object become intuitable? Solely in that its existence is posited in an intuitable form, i.e. in a form possible for intuition; but an object-form possible for intuition is actual only in virtue of an original intuiting that forms the object.

For the original intuiting, the object itself is reflected as image, and the image as object. In the form of the image, however, the object-reflection is superficial; the essential content has its exhaustive form in the concept. For our knowledge it is the object that determines the concept; but in the matter itself it is the concept that determines the object. In order to determine the object, the concept must beforehand be determined by a comprehending independent of our concept, hence by a knowledge independent of our knowledge.

The intuiting, thinking, and comprehending of omniscience, in which existence should have its original ground, is indeed incomprehensible from the standpoint of finitude (simultaneous and successive knowledge; intuitive and discursive knowledge;⁴ knowledge's relative and absolute limit); but as omniscience in represented form is incomprehensible to finite knowledge, finite knowledge becomes at the same time incomprehensible to itself: in order to comprehend itself, finite knowledge must recognize its dissolution in the infinite.

⁴Cf. Grundrid. Log. I. S. 230–63.

B. Power.

C. Truth.

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